

## BIRSK, Russian Federation

WMO: 286210

Lat: 55.4050N

Lon: 55.5517E

Elev: 186

StdP: 99.11

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -28.0      | -24.4 | -31.4                         | 0.2 | -27.9 | -27.5 | 0.3 | -24.0 | 11.8                  | -3.9 | 8.8 | -7.4 | 2.0                   | 20   | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.8                    | 30.2            | 20.2 | 28.7 | 19.5 | 27.0 | 18.7 | 21.6                | 27.9 | 20.6 | 26.9 | 19.6 | 25.3 | 2.7                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.4                            | 14.5 | 25.1 | 18.3 | 13.5 | 23.9 | 17.4 | 12.7 | 23.1 | 63.8          | 27.9 | 60.2 | 26.8 | 56.7 | 25.4 | 25.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.6           | 6.5  | 5.7 | DB                         | -30.3 | 32.6               | 3.8 | 2.0                                                | -33.0 | 34.0       | -35.2 | 35.2       | -37.4 | 36.3       | -40.1 | 37.7 | (4) |
| (5)               |      |     | WB                         | -30.2 | 22.7               | 3.8 | 1.5                                                | -32.9 | 23.8       | -35.1 | 24.6       | -37.2 | 25.5       | -40.0 | 26.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 4.3    | -12.1 | -10.8 | -4.0  | 5.4   | 13.6  | 17.8  | 19.9  | 18.0  | 11.9  | 4.7   | -3.7  | -9.7  | (6)  |  |
|                                                                        | DBStd       | 12.69  | 6.70  | 6.51  | 4.61  | 5.75  | 5.40  | 4.57  | 3.88  | 4.28  | 4.72  | 4.62  | 5.92  | 6.89  | (7)  |  |
|                                                                        | HDD10.0     | 3122   | 684   | 582   | 434   | 164   | 26    | 3     | 0     | 1     | 32    | 174   | 411   | 611   | (8)  |  |
|                                                                        | HDD18.3     | 5307   | 943   | 815   | 692   | 389   | 164   | 64    | 29    | 60    | 198   | 421   | 661   | 869   | (9)  |  |
|                                                                        | CDD10.0     | 1058   | 0     | 0     | 0     | 25    | 138   | 237   | 308   | 248   | 90    | 11    | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 201    | 0     | 0     | 0     | 1     | 17    | 48    | 79    | 49    | 6     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1531   | 0     | 0     | 0     | 6     | 138   | 371   | 615   | 366   | 36    | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 419    | 0     | 0     | 0     | 1     | 24    | 99    | 178   | 112   | 5     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.7    | 2.9   | 2.9   | 3.0   | 2.7   | 2.8   | 2.5   | 2.2   | 2.2   | 2.4   | 2.8   | 2.8   | 2.9   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 556    | 39    | 33    | 32    | 29    | 43    | 65    | 53    | 52    | 49    | 62    | 52    | 48    | (15) |  |
|                                                                        | PrecMax     | 742    | 73    | 68    | 62    | 68    | 80    | 138   | 128   | 117   | 113   | 123   | 118   | 94    | (16) |  |
|                                                                        | PrecMin     | 373    | 6     | 7     | 3     | 0     | 6     | 7     | 14    | 7     | 6     | 9     | 8     | 11    | (17) |  |
|                                                                        | PrecStd     | 105    | 17    | 15    | 17    | 19    | 22    | 35    | 33    | 26    | 26    | 32    | 27    | 20    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 1.1   | 1.9   | 8.1   | 23.8  | 29.2  | 31.8  | 32.5  | 32.6  | 27.2  | 18.9  | 9.6   | 1.9   | (19) |  |
|                                                                        |             | MCWB   | 0.3   | 0.3   | 4.1   | 14.0  | 17.3  | 20.4  | 20.5  | 20.5  | 16.0  | 11.8  | 7.8   | 0.5   | (20) |  |
|                                                                        | 2%          | DB     | -0.2  | 0.6   | 5.2   | 20.2  | 26.9  | 29.4  | 30.2  | 29.6  | 23.9  | 16.0  | 6.6   | 1.0   | (21) |  |
|                                                                        |             | MCWB   | -0.9  | -0.4  | 2.2   | 12.0  | 16.6  | 19.7  | 20.8  | 20.0  | 15.1  | 10.1  | 5.3   | 0.3   | (22) |  |
|                                                                        | 5%          | DB     | -1.4  | -0.5  | 3.3   | 17.5  | 24.7  | 27.4  | 28.7  | 27.2  | 21.3  | 13.5  | 4.6   | 0.2   | (23) |  |
|                                                                        |             | MCWB   | -2.0  | -1.3  | 0.9   | 10.5  | 15.4  | 18.8  | 20.3  | 18.8  | 14.5  | 9.2   | 3.5   | -0.4  | (24) |  |
|                                                                        | 10%         | DB     | -2.9  | -2.1  | 1.8   | 14.3  | 22.4  | 25.3  | 27.0  | 24.9  | 19.0  | 11.3  | 2.9   | -1.0  | (25) |  |
|                                                                        |             | MCWB   | -3.6  | -2.9  | 0.1   | 8.7   | 14.1  | 18.0  | 19.6  | 18.1  | 13.4  | 8.2   | 1.8   | -1.7  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 0.6   | 0.7   | 4.5   | 14.8  | 19.1  | 23.0  | 23.2  | 22.6  | 18.0  | 13.5  | 8.0   | 1.2   | (27) |  |
|                                                                        |             | MCDB   | 1.1   | 1.4   | 7.4   | 23.0  | 27.0  | 30.1  | 29.1  | 28.4  | 23.2  | 16.5  | 9.4   | 1.5   | (28) |  |
|                                                                        | 2%          | WB     | -0.7  | -0.2  | 2.5   | 12.5  | 17.2  | 20.8  | 21.9  | 20.7  | 16.3  | 11.3  | 5.5   | 0.4   | (29) |  |
|                                                                        |             | MCDB   | -0.3  | 0.5   | 4.7   | 19.3  | 25.0  | 27.1  | 28.1  | 27.7  | 21.2  | 14.4  | 6.5   | 0.8   | (30) |  |
|                                                                        | 5%          | WB     | -2.0  | -1.2  | 1.3   | 10.8  | 16.0  | 19.6  | 21.1  | 19.6  | 15.0  | 9.7   | 3.6   | -0.4  | (31) |  |
|                                                                        |             | MCDB   | -1.5  | -0.5  | 2.8   | 17.0  | 23.5  | 25.4  | 27.4  | 26.1  | 20.2  | 12.7  | 4.5   | 0.2   | (32) |  |
|                                                                        | 10%         | WB     | -3.6  | -2.9  | 0.3   | 8.9   | 14.8  | 18.6  | 20.1  | 18.5  | 14.0  | 8.4   | 2.0   | -1.6  | (33) |  |
|                                                                        |             | MCDB   | -2.9  | -2.2  | 1.6   | 14.2  | 21.5  | 24.2  | 25.6  | 23.9  | 18.3  | 11.0  | 2.6   | -0.9  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.3   | 5.9   | 6.2   | 7.5   | 9.2   | 8.8   | 8.8   | 8.3   | 7.3   | 5.2   | 4.1   | 4.9   | (35) |  |
|                                                                        |             | MCDBR  | 5.0   | 5.0   | 6.6   | 10.9  | 11.8  | 11.4  | 11.0  | 11.1  | 10.4  | 8.8   | 4.5   | 4.5   | (36) |  |
|                                                                        |             | MCWBR  | 5.0   | 4.6   | 4.5   | 6.0   | 5.6   | 5.1   | 4.4   | 4.5   | 5.0   | 5.0   | 3.9   | 4.4   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 5.3   | 4.8   | 5.5   | 10.7  | 11.0  | 10.3  | 9.9   | 10.3  | 9.3   | 7.8   | 4.4   | 4.4   | (38) |  |
|                                                                        |             | MCWBR  | 5.3   | 4.7   | 4.1   | 6.3   | 5.8   | 5.3   | 4.7   | 4.8   | 5.1   | 5.3   | 4.1   | 4.3   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.294 | 0.318 | 0.370 | 0.408 | 0.396 | 0.390 | 0.408 | 0.427 | 0.379 | 0.354 | 0.315 | 0.293 | (40) |  |
|                                                                        | taud        |        | 2.031 | 2.025 | 1.950 | 2.081 | 2.268 | 2.341 | 2.296 | 2.224 | 2.326 | 2.274 | 2.158 | 1.999 | (41) |  |
|                                                                        | Ebn at Noon |        | 581   | 721   | 769   | 805   | 845   | 855   | 829   | 782   | 773   | 688   | 563   | 474   | (42) |  |
|                                                                        | Edh at Noon |        | 68    | 100   | 138   | 140   | 124   | 117   | 120   | 121   | 96    | 79    | 60    | 56    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.80  | 1.80  | 3.25  | 4.39  | 5.39  | 5.83  | 5.78  | 4.50  | 2.81  | 1.26  | 0.57  | 0.46  | (44) |  |
|                                                                        | RadStd      |        | 0.09  | 0.18  | 0.29  | 0.47  | 0.48  | 0.55  | 0.56  | 0.42  | 0.28  | 0.28  | 0.13  | 0.10  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page